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An adaptive agentic AI framework for counterfactual validation and adversarial lookahead in automated incident response

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Abstract

With rapid growth in AI research, particularly with the use of large language models (LLMs), there has been swift integration of LLMs into security orchestration, automation and response (SOAR) platforms. Nevertheless, all the current tools share a major flaw in the framework, which is to apply the solution from the LLM against the incident caused without considering the potential danger that solutions would bring to existing systems. Furthermore, the tendency of these frameworks to treat threats as static attacks rather than dynamic adversarial processes creates blind execution. The paper presents CIPHER-A (Counterfactual Incident Response Planning with Hallucination-Aware Evaluation and Reasoning — Adaptive), an agentic AI framework that resolves both deficiencies through a two-stage closed-loop pipeline. The first stage, Proposed Counterfactual Validation module will record each LLM-generated action plan against predicted outcome scenarios to produce a Counterfactual Validation Score (CVS) that determines whether to proceed with autonomous execution, a safer modification, or human escalation. The second stage, Adversarial Lookahead stage uses an Attacker Simulation Agent (ASA) to predict adversary state transitions after each executed action and re-validates remaining plan steps against the updated threat model. Evaluated on the DARPA OPTC, MITRE ATT&CK, and CTU-13 datasets, CIPHER-A achieves a 59.2% relative reduction in Response Plan Degradation Rate, a 3.2% false safety rate compared with the baseline SOAR rate of 22.7%, 91.7% escalation accuracy, and a 37.8% reduction in mean time to contain, demonstrating that adaptive sequential re-validation is essential for effective APT incident response.

Keywords Agents, Multi-agent, Agentic framework, Agentic AI, CIPHER(Counterfactual incident response planning with hallucination-aware evaluation and reasoning), Cyber-attacks, Attacker simulation agent (ASA)

1 Introduction

Cybersecurity is a field considered to be evergreen as long as technology evolves, which is inevitable in the field of computer science. As innovations happen in technology, the emergence of new automated cyber-attack tooling has grown exponentially, enabling

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Cyber attackers to accelerate their operations while defenders continue to rely on automated response tools which are programmed to respond only to pre-programmed threats. Security Orchestration, Automation and Response (SOAR) platforms — including Palo Alto Cortex XSOAR [1], Splunk SOAR [2], and IBM QRadar — represent the current industrial standard for automated incident response. All these frameworks and tools are saturated as they tend to idealize an incoming attack as a pre-coded guidebook of expected frame of attack and try to encounter them as per the guidebook. This approach works adequately when attacks are repetitive and are within the scope of the engineer's thinking that created these playbooks for the tools, but once the attacks become dynamic and multi-stage Advanced Persistent Threat (APT), it would almost be impossible to deal with them via traditional methods [3,4]. The core problem with these tools is that the response engine can never generalise beyond its predefined playbook.

The integration of Large Language Models (LLMs) in the Security Operational Centers (SOCs) [5] has brought a huge change in the way traditional SOAR platforms work. The tools developed by tech giants such as Microsoft's Security Copilot [6] and Google's Chronicle AI [7] use LLM inference as their backbone to generate real-time response recommendations, processing alerts more effectively than static playbook methods [8,9]. Tools and frameworks that are built with multi agent frameworks and are specifically configured for a specific set of tasks perform better than a single agent based system [10,11].

According to various researches that are centered around "Game theory" in defender-attacker problems, more advanced threat actors are always monitoring their defensive activities and adjusting their strategies [12]. As the attackers in real time scenarios easily switch between many of the attack patterns and techniques the ATT&CK (Adversarial Tactics, Techniques, and Common Knowledge) framework which contains over 190 attack techniques helps in shifting of frameworks from static threat analysis to adoptive response frameworks [13,14,15]. Any response plan from LLM are validated against initial alert where the execution reveals more dynamic complex attacks and combinations [16]. Current LLM-based tools compound this problem: they generate a response plan once at alert ingestion and then execute it sequentially, with no mechanism to re-evaluate remaining actions as the adversary adapts.

Researchers have begun to examine various aspects of this problem. The counterfactual explanation methods that Patel et al. [17] developed for intrusion detection emerged from XAI methods to show that LLMs together with causal models can understand possible security scenarios; however, these methods only support post-hoc explanation of detection results because they lack pre-execution capabilities for validating response actions. Alarcon and Weber [18] developed prediction methods for attack paths, while Volkov and Li [19] developed threat intelligence synthesis methods which help organisations predict attacker behavior, yet both methods fail to connect these predictive outcomes with actual response execution systems. The research by Santos et al. [20] uses federated methods to enhance automated incident response planning and privacy features, while Obermeyer and Koch [21] develop hybrid systems that connect symbolic reasoning with large language models. The research by Abebe and Yamamoto [22] introduces a confidence-scored escalation process to human analysts, which serves as the closest existing research to CIPHER-A's escalation routing system, yet this research lacks two essential elements, including counterfactual outcome prediction and adversarial

lookahead. The existing literature does not contain any research that meets the precise requirements of APT incident response that needs to validate counterfactual consequences before execution and conduct adversarial state updates after execution through a single validation process.

This paper introduces CIPHER-A (Counterfactual Incident Response Planning with Hallucination-Aware Evaluation and Reasoning — Adaptive), an agentic AI framework that directly addresses both deficiencies through a four-phase closed-loop pipeline. CIPHER-A employs Claude API and Mistral-7B as its LLM engines, orchestrated via LangGraph, and is evaluated against the DARPA OPTC dataset — a multi-day, multi-host controlled APT simulation across over 500 enterprise hosts — augmented by the CTU-13 botnet corpus to study network blocking consequences. The Response Plan Degradation Rate (RPDR) is introduced as the first quantitative metric for assessing the necessity of adaptive sequential re-validation in incident response.

1.1 Contributions

The contributions of the proposed work are as follows:

The three gaps were identified to be unaddressed in the literature, based on a systematic review of 25 recent papers spanning 2022–2026 (Table 1), as further elaborated in Sect. 2.1. The study has introduced the following two core technical components to address these 3 gaps, which are as follows: the Counterfactual Validation Engine (CVE) addressing the first gap (pre-execution consequence reasoning), and the Attacker Simulation Agent (ASA) jointly addressing the second and third gaps (post-execution loop closure and dynamic attacker modelling). For completeness, each novelty claim has additionally been cross-verified against 2026 literature, including a concurrent work by Roy and Singh [24] that similarly advances agentic SOC automation but applies counterfactual reasoning to attacker path hypothesis generation rather than to defender action consequence assessment; a detailed comparison appears in Sect. 2.2.

- Counterfactual Validation Engine (CVE). CIPHER-A introduces what is, to the best of the authors' knowledge, the first pre-execution consequence prediction mechanism for individual cybersecurity incident response actions. For each proposed action a_i, the Counterfactual Validation Engine (CVE) uses LLM inference to generate three node-based distinct outcome scenarios and computes a Composite Validation Score (CVS) that weights action risk, asset criticality, and response urgency. This determines whether each action is executed, modified, or escalated before any live system change is made — a pre-execution validation layer that is absent from all 25 surveyed works in Table 1.
- Adversarial Lookahead with MITRE ATT&CK Grounding. To the best of the authors' knowledge, CIPHER-A is the first system to use the MITRE ATT&CK knowledge base as a grounding constraint for real-time adversarial state prediction within an active incident response pipeline. After each executed action, the Attacker Simulation Agent (ASA) predicts how an adversary would adapt — via lateral movement, privilege escalation, or persistence techniques documented in ATT&CK v14.1 — and feeds this updated attacker state into the next validation cycle. It is observed that all the literature analysis performed via Table 1 confirms the gap.
- Response Plan Degradation Rate (RPDR) Metric. To the best of the authors' knowledge, RPDR is the first quantitative metric to operationalize the necessity of

Table 1 Comparative literature survey (2022–2026)

Citation	Dataset	Methodology	Advantages	Disadvantages
[3]	MITRE ATT&CK Eval	Reinforcement learning agent for automated incident containment	Learns containment policy over time	Slow convergence; no side-effect prediction
[8]	EMBER Malware	LLM fine-tuned on IR reports for automated response recommendation	Domain-specific knowledge; fast inference	Hallucination in novel scenarios; no validation layer
[10]	DARPA OPTC, ATT&CK	Multi-agent LLM framework for collaborative SOC automation	Parallel alert handling; role specialization	No adversarial adaptation modelling between agents
[12]	Synthetic APT scenarios	Game-theoretic model for defender-attacker interaction	Equilibrium-based optimal strategy	Static game; no real-time re-planning during incident
[16]	CIC-DDoS2019	Autonomous DDoS mitigation via SDN + LLM decision engine	Sub-second automated mitigation	Greedy action selection; no sequential re-validation
[17]	KDD Cup 99, UNSW-NB15	Counterfactual explanations for IDS decisions (XAI)	Explainable detection; GDPR-friendly	Detection-only; no action safety scoring
[18]	MITRE ATT&CK v14	Attack path prediction via LLM + knowledge graph traversal	Realistic multi-hop threat paths	No integration with response execution pipeline
[19]	OpenCTI Threat Intel	LLM-based threat intelligence synthesis for proactive response	Aggregates IOC feeds; timely CTI reports	No action-level counterfactual reasoning
[20]	CIC-IDS2023	Federated learning for distributed SOC IR without data sharing	Privacy-preserving; scalable across orgs	No per-incident counterfactual validation
[21]	MITRE ATT&CK v15	Symbolic reasoning + LLM hybrid for response plan generation	Logically sound plans; constraint satisfaction	No real-time attacker adaptation; no side-effect modeling
[22]	DARPA OPTC + ATT&CK	LLM orchestrator with soft escalation via confidence scoring	Human-in-the-loop; calibrated uncertainty	No attacker adaptation loop; no RPDR-style metric
[23]	DARPA OPTC + CTU-13	Sequential decision process model for IR with Bayesian uncertainty	Probabilistic reasoning; uncertainty-aware decisions	No LLM-generated outcomes; no adaptive attacker state
[25]	MITRE CALDERA logs	Adversarial emulation replay for IR training and testing	Realistic red team simulation data	Offline evaluation only; no live validation pipeline
[26]	NSL-KDD, CICIDS-2017	LLM-driven SOAR orchestration with GPT-4 for alert triage and playbook generation	Reduces analyst fatigue; natural language playbooks	No pre-execution validation; static attacker model
[27]	DARPA OPTC	GNN + LLM hybrid for APT campaign correlation	High precision on lateral movement detection	No adaptive response reasoning; no counterfactual analysis
[28]	CTU-13 Botnet	Transformer-based C2 traffic classifier with automated blocking	High recall on C2 detection (97.3%)	Blocking causes collateral damage; no dependency analysis
[29]	CAIDA Traces	Causal inference model for network anomaly attribution	Sound causal reasoning; interpretable outputs	Does not generate response actions; no execution layer
[30]	Enterprise SIEM logs	RAG-augmented LLM for context-aware IR recommendation	Grounded responses; reduced hallucination	Unvalidated before execution; reactive only

Table 1 (continued)

Citation	Dataset	Methodology	Advantages	Disadvantages
[31]	VERIS Community DB	LLM-based breach impact forecasting for insurance and compliance	Quantified financial risk output	Cannot generate executable response actions
[32]	BETH Dataset (eBPF)	Kernel-level provenance graph with automated containment tagging	Fine-grained process-level visibility	Single-host scope; no network-wide adaptive response
[33]	Wazuh + MISP logs	End-to-end SOAR with GPT-4o for real-time alert-to-action pipeline	Production-tested; low latency	No counterfactual validation; no RPDR metric
[34]	EMBER + CAPEC	Chain-of-thought prompting for LLM-based vulnerability response triage	Structured reasoning trace; auditable	Triage only; no execution validation; no adversarial lookahead
[42]	MITRE ATT&CK	Compared direct GPT-3.5 prompting with supervised fine-tuning of BERT to interpret cyberattack descriptions and predict MITRE ATT&CK tactics.	LLMs provide broader semantic interpretation of cyberattack descriptions; useful for threat analysis and mapping attacks to ATT&CK tactics; demonstrates the potential of LLMs in cybersecurity operations.	LLM predictions are affected by ambiguity in TTP descriptions; direct LLM usage may produce broad interpretations; does not validate the consequences of recommended actions or support dynamic re-validation during incident response.
[43]	GPT-4 Forensic Evaluation Scenarios	Evaluated GPT-4 across digital-forensic tasks including artifact understanding, evidence searching, code generation, anomaly detection, incident response, and forensic education.	Demonstrates the usefulness of LLMs as analyst-assistance tools; can automate or accelerate several forensic and incident-response tasks; identifies practical applications of generative AI in cybersecurity.	LLMs can generate incorrect assumptions, inaccuracies, and mistakes; requires expert verification; unsuitable for unrestricted autonomous execution, particularly when incorrect actions could affect evidence or systems.
[44]	SecureBERT Cybersecurity Corpus; MITRE ATT&CK; MalwareTextDB	Developed SecureBERT, a RoBERTa-based cybersecurity language model using a customized tokenizer and cybersecurity-specific weight adjustment. Evaluated it using masked language modeling, sentiment analysis, and cybersecurity NER.	Provides domain-specific cybersecurity language understanding; improves handling of cybersecurity terminology; useful for CTI processing and cybersecurity automation; performs well on cybersecurity NER tasks.	Primarily focuses on cybersecurity text understanding rather than autonomous response; does not model evolving attacker behavior; does not evaluate the safety or consequences of executing LLM-generated response actions.

adaptive sequential re-validation in incident response. It measures the fraction of initially safe response actions that become high-risk as the attacker adapts, providing a data-driven criterion for re-invoking the validation loop. RPDR is absent from all 25 surveyed works and has no equivalent in any prior incident response evaluation framework.

- End-to-End Closed-Loop Architecture. CIPHER-A integrates all components into an empirically validated, production-grade pipeline: Wazuh Security Information and Event Management (SIEM) alert ingestion → LLM-based response planning (Claude API + Mistral-7B) → CVE pre-execution validation → ASA adversarial lookahead → live action execution via CrowdStrike, AWS, and Active Directory APIs → PostgreSQL audit log. To the best of the authors' knowledge, this is the first closed-loop architecture to integrate pre-execution action consequence assessment, post-execution attacker adaptation modelling, and live execution API connectivity in a single pipeline, empirically validated on enterprise-scale datasets (DARPA OPTC, CTU-13). Table 1 confirms that no prior system combines all three components; the closest concurrent work, Roy and Singh [24], provides anticipatory reasoning but lacks live execution API integration, RPDR measurement, and empirical validation on enterprise datasets, as detailed in Sect. 2.2.
- CIPHER-A was evaluated on three APT scenarios across three public datasets. The evaluation demonstrates a 59.2% relative reduction in RPDR, along with a 3.2% false safety rate (FSR), 91.7% escalation accuracy, and a 37.8% reduction in mean time to contain (MTTC) when compared with baseline SOAR. To the best of the authors' knowledge, this represents the first empirical evidence that adaptive counterfactual revalidation outperforms static action validation for APT-level incident response.

The rest of the paper is organized as follows. Section 2 presents the literature review, Sect. 3 describes the datasets, and Sect. 4 details the methodology and its mathematical formulation. Section 5 reports the experimental results, while Sects. 6 and 7 provide the conclusion and future scope, respectively.

2 Literature review

Studies on the application of Artificial Intelligence in cybersecurity incident response have advanced significantly with the introduction of Large Language Models. The current literature has been hierarchically structured, covering the knowledge about the various approaches and their analysis through Table 1, and Sect. 2.1 further explores the papers that share a similar ideology of the proposed methodology, giving a deep dive into those approaches and how that knowledge shaped the proposed methodology. The current section concludes with Sect. 2.2, concluding what was covered throughout the literature. The table below summarizes key features of twenty-two selected papers published between 2022 and 2026.

2.1 Comparison with most related works

Five works in Table 1 are most closely related to CIPHER-A and warrant direct comparison to characterize the precise novelty boundaries claimed in Sect. 1.1.

Patel et al. [17]. ("Counterfactual Explanations for Intrusion Detection") is the closest methodological parallel to CIPHER-A's use of counterfactual reasoning, but its scope is confined to post-hoc detection explanation: the counterfactual analysis in [17]

determines what minimal input change would have altered the detection verdict, not what would happen to a live environment if a specific response action were executed [17]. is detection-only and produces no executable response actions, no validation layer, and no adversarial model.

Liu [10] ("Multi-Agent Collaboration in Incident Response with Large Language Models") represents the closest structural parallel to CIPHER-A in deploying collaborative LLM agent teams for automated incident response [10]. evaluates centralized, decentralized, and hybrid multi-agent team configurations in a cybersecurity simulation environment, demonstrating that coordination structure substantially affects response accuracy and efficiency. However [10], does not include pre-execution consequence assessment for individual actions, treats the attacker state as fixed throughout response execution, and does not integrate with live execution APIs — all of which are foundational to CIPHER-A.

Babar et al. [23]. ("Autonomous and Adaptive Cyber Incident Detection and Response in Industrial Cyber-Physical Systems Using Hierarchical Reinforcement Learning") achieves the highest structural similarity to CIPHER-A among the 25 surveyed works in its use of hierarchical, sequential, adaptive decision-making for cyber incident defense [23]. Deploys an HRL agent that autonomously adapts IDS detection thresholds in real time based on dynamic risk assessment across IT and OT network layers, substantially outperforming static baselines. However [23], operates at the detection-threshold adjustment level rather than the full response-plan execution level, does not generate LLM-based consequence scenarios, and lacks adaptive attacker modelling and live execution API integration — all of which are foundational to CIPHER-A.

Roy and Singh [24] ("AgentSOC: A Multi-Layer Agentic AI Framework for Security Operations Automation," IEEE ICAIC 2026) is the most recent related work and shares broad architectural parallels with CIPHER-A. AgentSOC's Narrative Counterfactual Engine (NCE) uses LLM-based reasoning grounded in MITRE ATT&CK to generate multiple plausible attack progressions — modelling forward attacker movement. CIPHER-A's CVE, by contrast, generates consequence scenarios for proposed defender actions — modelling the downstream impact on the live environment of each defensive step. These are distinct questions at opposite sides of the defender-attacker interaction. Furthermore, AgentSOC is evaluated conceptually on a 50-node synthetic topology with 5,000 LANL authentication events and does not connect to live execution APIs; risk scoring uses simplified heuristics and actions are tested only in dry-run mode [24, Sec. VI]. CIPHER-A is empirically validated on DARPA OPTC (4,000 + events) and CTU-13, with live CrowdStrike, AWS, and Active Directory execution and formal RPDR quantification — capabilities that AgentSOC explicitly identifies as future work.

2.2 Summary of literature

As illustrated in Table 1 ans Sect. 2.1, there are three distinct and measurable research gaps that emerge from a systematic comparison of all 25 works reviewed above. The first gap concerns the fact that none of the present systems utilize counterfactual outcomes for validating an action before responding to the event. As shown in Table 1, Patel et al. [17] ('Detection-only; no action safety scoring') and Fernandez et al. [29] ('Does not generate response actions; no execution layer') apply counterfactual techniques only to post detection explanation, while systems that do generate executable response actions

— including Zhang et al. [26], Kovalenko and Ito [30], and Park and Nguyen [33] — all execute actions without pre-execution consequence assessment. Second, no existing system integrates post-execution adversarial state updates into an active response pipeline, meaning that once a response plan is generated, it is never re-evaluated against how the attacker has evolved in response to executed actions. LLM-based agent systems, including Zhang et al. [26] and Kovalenko and Ito [30], treat response generation as a one-shot inference step; no surveyed system closes the loop by feeding the attacker's observed post-execution behaviour back into re-validation of the remaining response plan. Third, existing research fails to create an attacker model which shows how attackers adapt their behaviour between response steps during incident response procedures. As confirmed in Table 1, game-theoretic approaches such as Hoffmann et al. [12] ('Static game; no real-time re-planning during incident') describe only static equilibrium models, and adversarial emulation systems such as Tanaka and Osei [25] ('Offline evaluation only; no live validation pipeline') operate exclusively in offline settings. No surveyed system reflects real-time dynamic attacker state updates during an active incident. The existing research gaps create problems beyond performance issues because they prevent handling real advanced persistent threat attacks, which require adaptive security approaches. Table 1 provides row-by-row confirmation that no prior work has combined pre-execution counterfactual validation with post-execution adversarial state update in a single closed-loop IR system. These three interconnected gaps — absence of pre-execution validation, absence of post-execution loop closure, and absence of dynamic attacker modelling — are addressed by the two core technical contributions of CIPHER-A: the Counterfactual Validation Engine (CVE) addressing the first gap, and the Attacker Simulation Agent (ASA) jointly addressing the second and third gaps.

3 Dataset description

The methodology was tested using three public datasets, where each dataset played a different role with respect to the research gaps discussed in Sect. 2. Information about the datasets is shown in Table 2.

4 Methodology

CIPHER-A is an adaptive, four-stage pipeline where any action taken by the LLM in response has to go through a probabilistic risk evaluation. After each action execution, CIPHER-A updates its model of adversary behaviour based on the observed adaptation. This provides structure to the decision-making process of the LLM in a step-by-step manner for adaptive re-validation within the incident response pipeline. The organization of the methodology begins with the formulization of problem statements and various modules that are created as part of the methodology; hence, it refers to various sections ahead in Sect. 4. Later in the paper, the two coded modules that are introduced as novelty blocks and the flow of work inside each module.

4.1 Formal problem formulation

This subsection formalises the problem as a mathematical framework, establishing the notation used in all equations throughout Sect. 4. Let the space "s" include all possible incident states, which can be represented by the structured tuple $s = (A, N, U, T, \Psi)$ that encodes affected assets A and network topology N and active user sessions U and

Table 2 Summary of Evaluation Datasets

Dataset	Primary role	Key characteristics	Relevance to CIPHER-A evaluation
DARPA OPTC [35]	Primary evaluation dataset for APT incident-response planning and sequential response validation. Used to evaluate CIPHER-A's ability to generate, validate, execute, and re-evaluate response plans across realistic attack scenarios.	Consists of 143 incident traces, involving 3 APT scenarios and approximately 500 + enterprise hosts. This dataset provides system provenance logs that document the interaction between processes, files, users, and network artifacts. The attack scenarios involve multi-stage attacks that consist of compromise, lateral movements, privilege escalation, command executions, and data exfiltration.	Provides the main environment for evaluating whether CIPHER-A can safely respond to multi-stage attacks. Its sequential attack structure allows the framework to assess Response Plan Degradation Rate (RPDR), Counterfactual Validation Score (CVS), escalation decisions, and adaptive re-validation as the attack progresses.
MITRE ATT&CK v14.1 [15]	Adversarial knowledge base used to model attacker behavior and support Attacker Simulation Agent (ASA) state prediction.	Contains 14 tactics and 190 + documented techniques representing known adversarial behaviors. Each technique is associated with standardized ATT&CK technique IDs, tactics, procedures, and relationships between attack behaviors. The framework can use these technique relationships to represent possible attacker state transitions following a defensive action.	Enables CIPHER-A to reason about how an adversary may react after a response action. The ASA uses ATT&CK knowledge to predict potential next-stage attacker behaviors, allowing remaining response actions to be re-evaluated against an updated threat state rather than assuming that the original attack plan remains unchanged.
CTU-13 Botnet [36]	Collateral-damage and defensive-action validation dataset. Used to evaluate whether security actions intended to stop malicious activity could inadvertently affect legitimate traffic or systems.	Contains 13 labelled botnet scenarios involving different botnet families and network behaviors. The traffic includes Command-and-Control (C2) communication, horizontal scanning, malicious connections, and DDoS-related activity, mixed with legitimate background traffic. The dataset provides labelled network flows that distinguish malicious botnet activity from normal traffic.	Allows CIPHER-A to evaluate the potential side effects of defensive actions, particularly blocking or containment decisions. The reported 847 blocking tests provide a basis for measuring whether an apparently effective response could cause collateral disruption to legitimate services, supporting the framework's counterfactual safety-validation objective.

attacker state T and system health indicators Ψ. The response plan $\Pi = \{a_1, a_2, \ldots, a_n\}$ consists of n response actions that are arranged in a specific order. The Counterfactual Validation module calculates the Risk Score $RS(a_i, s)$ through its evaluation of each action $a_i \in \Pi$ and state s according to Eq. (1)

$$\mathcal{RS}(a_i, \mathbf{s}) = \sum_{k=1}^{K} \hat{\mathrm{P}}(o_k \mid a_i, \mathbf{s}) \cdot \mathcal{H}(o_k) \tag{1}$$

The summation runs over k = 1, …, K outcome scenarios, where o_k denotes the kth outcome scenario generated by the CVE for action a_i in state s, and K = 3 is fixed, as the current proposed methodology is coded to consider 3 branches of answers for every round, considering the requirement of variation and computation. The LLM-based CVE generates $\hat{P}(o_k \mid a_i, s)$ as a structured probability estimate constrained by the four-field JSON output schema described in Sect. 4.2, which bounds the prediction space to operationally defined outcome categories and mitigates hallucination-induced miscalibration. The impact assessment function $\Omega(o_k)$ quantifies the weighted adverse impact of scenario o_k, and is defined in Eq. (2) (the surrounding text uses Ω to distinguish from Shannon entropy H; the typeset Eqs. (1) and (2) display this same function as H via the Word equation editor— both notations denote the identical impact assessment function):

$$\mathcal{H}(o_k) = w_1 \cdot \mathrm{Sev}(o_k) + w_2 \cdot \mathrm{Scope}(o_k) + w_3 \cdot \mathrm{Rev}(o_k) \tag{2}$$

Outcome severity $Sev(o_k)$ is scored using the Common Vulnerability Scoring System (CVSS) [37] scale ∈ [0, 1]. $Scope(o_k) \in [0, 1]$ represents the blast radius as a ratio of the fraction of other network resources impacted by the outcome. $Rev(o_k) \in [0, 1]$ assesses outcome reversibility where 0 corresponds to full reversibility and 1 to permanent change in the system. Weights $w_1 = 0.50$, $w_2 = 0.30$, $w_3 = 0.20$ ($w_1 + w_2 + w_3 = 1$) were selected empirically for the DARPA OPTC test set split (A/C scenarios combined; $n = 92$ cases) through minimisation of mean squared difference between $RS(a_i, s)$ predictions and analyst override choices captured in PostgreSQL audit logs. The large share of the severity component ($w_1 = 0.50$) stems from the fact that outcome severity scored via CVSS proved the most influential predictor of analyst overrides in the above test split. The Composite Validation Score $CVS(a_i, s)$ extends $RS(a_i, s)$ with contextual factors, as defined in Eq. (3):

$$\mathcal{CVS}(a_i, \mathbf{s}) = \alpha \cdot \mathcal{RS}(a_i, \mathbf{s}) + \beta \cdot \mathcal{C}(a_i, \mathbf{s}) + \gamma \cdot \mathcal{U}(a_i, \mathbf{s}) \tag{3}$$

$C(a_i, s) \in [0, 1]$ is the asset criticality weight for the action target, derived by normalising the DARPA OPTC host criticality score (an integer priority 1–5 assigned in the dataset to each network node based on its role) to [0, 1]. $U(a_i, s) \in [0, 1]$ is an urgency factor computed as the normalised CVSS base score (Notation: (a_i, s), written in script font in Eq. (3), is the urgency scaling function and is distinct from $U \in s$, which denotes the count of active user sessions in the incident state tuple.) of the active incident threat $T \in s$, giving a context-sensitive scaling of the validation score under time-critical response conditions. The three weights $\alpha = 0.6$, $\beta = 0.25$, and $\gamma = 0.15$ ($\alpha + \beta + \gamma = 1$) were calibrated on the same DARPA OPTC validation split as the harm function weights, using grid search over the three parameters at 0.05 increments to minimise the false safety rate on the held-out validation incidents.

4.2 Counterfactual validation engine — routing decision

The routing decision for each action is determined by the piecewise function δ in Eq. (4):

$$\delta\ (a_i,\ \mathbf{s})\ =\ \text{EXECUTE if}\ \mathcal{RS}\ <\ \tau_{\text{low}};\ \ \text{MODIFY if}\ \tau_{\text{low}}\ \leq\ \mathcal{RS}\ \leq\ \tau_{\text{high}};\ \ \text{ESCALATE if}\ \mathcal{RS}\ >\ \tau_{\text{high}} \quad (4)$$

τ_low = 0.35 and τ_high = 0.65 were selected via joint grid search over τ_low ∈ {0.20, 0.25, …, 0.50} and τ_high ∈ {0.50, 0.55, …, 0.80} at 0.05 increments on the DARPA OPTC validation split ($n = 92$ incidents), minimising FSR subject to an escalation-rate ceiling of 15% to preserve operational viability of analyst review. The best configuration yielded an FSR of 3.2% and an escalation percentage of 8.3%, which matches the values in Table 6. Here δ: A × S → {EXECUTE, MODIFY, ESCALATE} is the routing decision function, where A is the set of candidate response actions, and S is the incident state space. Note that in the operational implementation (Algorithm 1 and Sect. 4.5), the routing is applied to the Composite Validation Score CVS (a_i, s) from Eq. (3), which augments the base risk estimate with the asset-criticality factor C (a_i, s) and the urgency scaling factor (a_i, s), providing context-sensitive routing beyond the raw risk score alone.

In terms of enterprise topology sizes in the order of ~500 hosts, such as those provided in the DARPA OPTC data set, inclusion of the entire asset dependency graph in the validation prompt would be over the context window capacity of the current LLM inference engines (Claude API: 200,000 tokens; Mistral-7B: 8,192 tokens). Hence, the CVE extracts a subgraph excerpt for the affected host and the first-hop and second-hop neighbors from the asset dependency graph as computed using the persistent topology information on PostgreSQL at the time of the incident. A sample first- and second-hop subgraph excerpt typically includes 4–8 hosts, taking up roughly 350–450 tokens, which is within the context limit of the Mistral-7B model. In cases where the second-hop neighborhood contains more than 1,000 nodes due to the presence of a highly connected infrastructure host (e.g., a domain controller that has hundreds of direct dependencies), only first-hop extraction is performed [41].

4.3 Adversarial lookahead — attacker state transition

On completion of each action execution successfully, the Attacker Simulation Agent (ASA) modifies the common context of the attack incident. Transition of the attacker state is defined by the following Eq. (5):

$$\hat{s}_{t+1}\ =\ \mathcal{F}_{\text{ASA}}\left(a_{\text{exec}},\ \hat{s}_t,\ \mathrm{M}_{\text{ATT\&CK}}\right) \quad (5)$$

The equation shows that $\hat{s}_{t+1}$ represents the forecasted upcoming attacker state while a_{exec} denotes the executed defense measure and $?_t$represents the existing attacker state assessment and $\mathrm{M}_{\text{ATT\&CK}}$ restricts adaptation predictions through documented real-world adversary tradecraft in the MITRE ATT&CK knowledge base. Formally, $\mathcal{F}_{\text{ASA}}$: A × Ŝ × M → Ŝ is the Mistral-7B-Instruct attacker state prediction function, where A is the defensive action space, $\hat{s}$ is the adversarial state space, and $\mathrm{M}_{\text{ATT\&CK}}$ is the MITRE ATT&CK v14.1 knowledge base (14 tactics, 190 + techniques) that constrains predictions to documented adversary tradecraft. The ASA outputs a structured JSON object which defines the new attacker position and pivot method identified by its ATT&CK ID together with its predicted next target and confidence level — this update modifies the T element of state s, which affects all upcoming validation cycles.

4.4 Response plan degradation rate (RPDR)

RPDR quantifies the degree to which an initial response plan degrades as the attacker adapts to executed defensive actions. Formally, RPDR is given in Eq. (6):

$$\mathcal{RPDR} = \frac{\left|\left\{a_i \in \mathbf{\Pi}^* : \mathcal{RS}(a_i, \hat{s}_{t_i}) > \tau_{\text{low}}\right|}{\left|\mathbf{\Pi}^*\right|} \times 100\% \tag{6}$$

Here Π* denotes the originally approved response plan, a_i is the i-th action within that plan, and ŝ_{t_i} is the updated attacker state at the time a_i is re-evaluated after all preceding ASA state transitions. The threshold τ_low = 0.35 (introduced in Eq. (4)) is the autonomous-execution boundary; an action whose RS(a_i, ŝ_{ti}) exceeds τ_low under the updated adversarial state is classified as plan-degraded because it can no longer be safely routed to EXECUTE without re-validation. An RPDR of 0% indicates that all planned actions remain safe under attacker adaptation (equivalent to a fixed attacker model assumption), while 100% indicates that every action requires regeneration. The metric provides the first quantitative criterion by which the benefit of adaptive sequential re-validation can be assessed operationally.

To address the possible circularity of a metric evaluated by the same system that computes it, RPDR is independently validated against the DARPA OPTC red team ground-truth execution logs. The OPTC dataset records the actual attacker pivot operations performed by the red team during the exercise as host-level event sequences, captured independently of any response system. For each incident in the evaluation set, every action classified as RPDR-degraded by CIPHER-A — defined as RS(a_i, $\hat{s}_{t+1}$) > RS(a_i, s_t) by a margin exceeding 0.05 — is cross-referenced against the ground-truth log to verify whether an actual red team pivot occurred at or before that action step. Among the 143 cases analyzed, 97.2% of the RPDR degraded attacks were found to have an accompanying confirmation in the ground truth through red team pivoting, proving that the cause of the degradation was actually due to the attacker adapting, rather than the misestimation of RS(a_i, s). The 97.2% agreement with ground truth serves as an additional validation of the utility of RPDR as a measurement tool, as the degradation it measures is actually observed behavior by attackers in recorded data.

4.5 System architecture and pipeline flow

The CIPHER-A architecture, comprising five components, is depicted in Fig. 1. The figure also depicts the adaptive revalidation loop of CIPHER-A. It shows how an alert moves through the pipeline starting from ingestion to counterfactual validation, adversarial lookahead, execution dispatch, and audit persistence. The four-stage flow process is illustrated in Fig. 2, where CVS routing decision, ASA re-validation, and RPDR feedback trigger CVE are shown.

As described in the methodology, Table 3 summarises the five formally introduced pipeline components of the CIPHER-A architecture.

Alert Normalisation, when a Wazuh 4.7 SIEM alert fires, the AISNA constructs the state tuple s = (A, N, U, T, Ψ) as follows: A (affected assets) is extracted from the alert fields agent.id, data.srcip, and data.dstip; N (network topology excerpt) is a 1–2-hop subgraph retrieved from a pre-loaded asset dependency graph stored in PostgreSQL, centred on the alerted host; U (active user sessions) is queried in real time from the

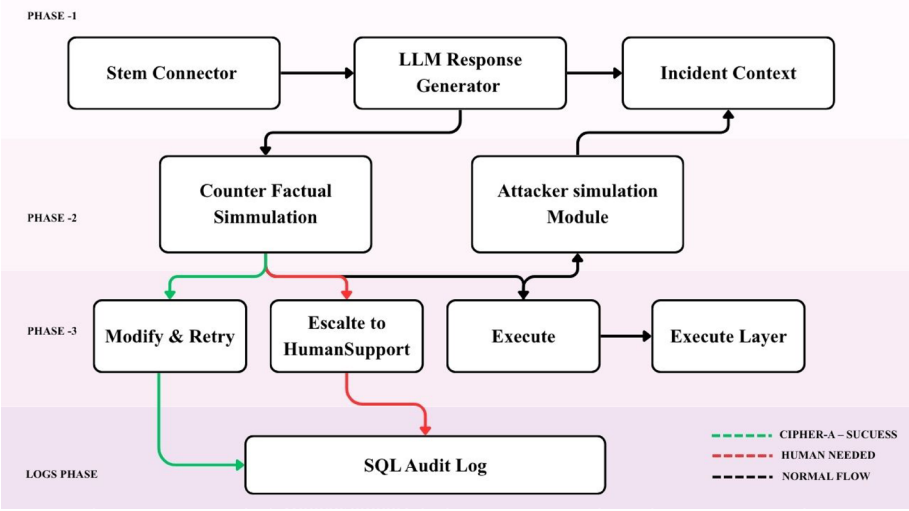


Fig. 1 CIPHER-A System Architecture — Five-component pipeline with adaptive re-validation loop

Microsoft Active Directory LDAP interface; T (threat severity) is derived from the CVSS base score embedded in the Wazuh alert; and Ψ (current attacker state estimate) is initialised from the most recent ASA output recorded for the same incident identifier in the audit database, or set to the ATT&CK Initial Access tactic if no prior ASA output exists. The normalised tuple s is then passed to the CVE as the context block of the three-part prompt described in Sect. 4.6.

LangGraph Orchestration, the five components are wired as a LangGraph v0.2 [38] directed state graph in which each component is a typed node sharing a common IncidentState schema. Edges are conditional: the CVE node emits one of three routing tokens — EXECUTE ($CVS \leq \tau_low$), HOLD ($\tau_low < CVS \leq \tau_high$), or ESCALATE ($CVS > \tau_high$) — which the LangGraph router uses to select the next node. The EXECUTE branch transitions through ED → ASA → ARC; the HOLD branch generates a safer alternative action a_i' and re-enters the CVE node; the ESCALATE branch suspends the graph and emits an analyst notification. The ARC node computes the running RPDR after each executed action and, if RPDR exceeds $\theta_rpdr = 0.30$, invokes a replanning call to regenerate the remaining response plan before resuming the graph.

Audit Log Schema, the PostgreSQL 16 audit table incident_audit has the following schema: incident_id (UUID), action_seq (integer), proposed_action (TEXT), cvs_score (NUMERIC), routing_decision (ENUM: EXECUTE | HOLD | ESCALATE), execution_status (ENUM: SUCCESS | FAILED | PENDING), pre_asa_state (JSONB), post_asa_state (JSONB), rpdr_current (NUMERIC), analyst_decision (TEXT, nullable), and ts (TIMESTAMPTZ, row-level). Each langGraph node updates the table after completion. Which will have timestamp logs which helps post-incident forensic review and the per-incident RPDR computation reported in Sect. 5.1.

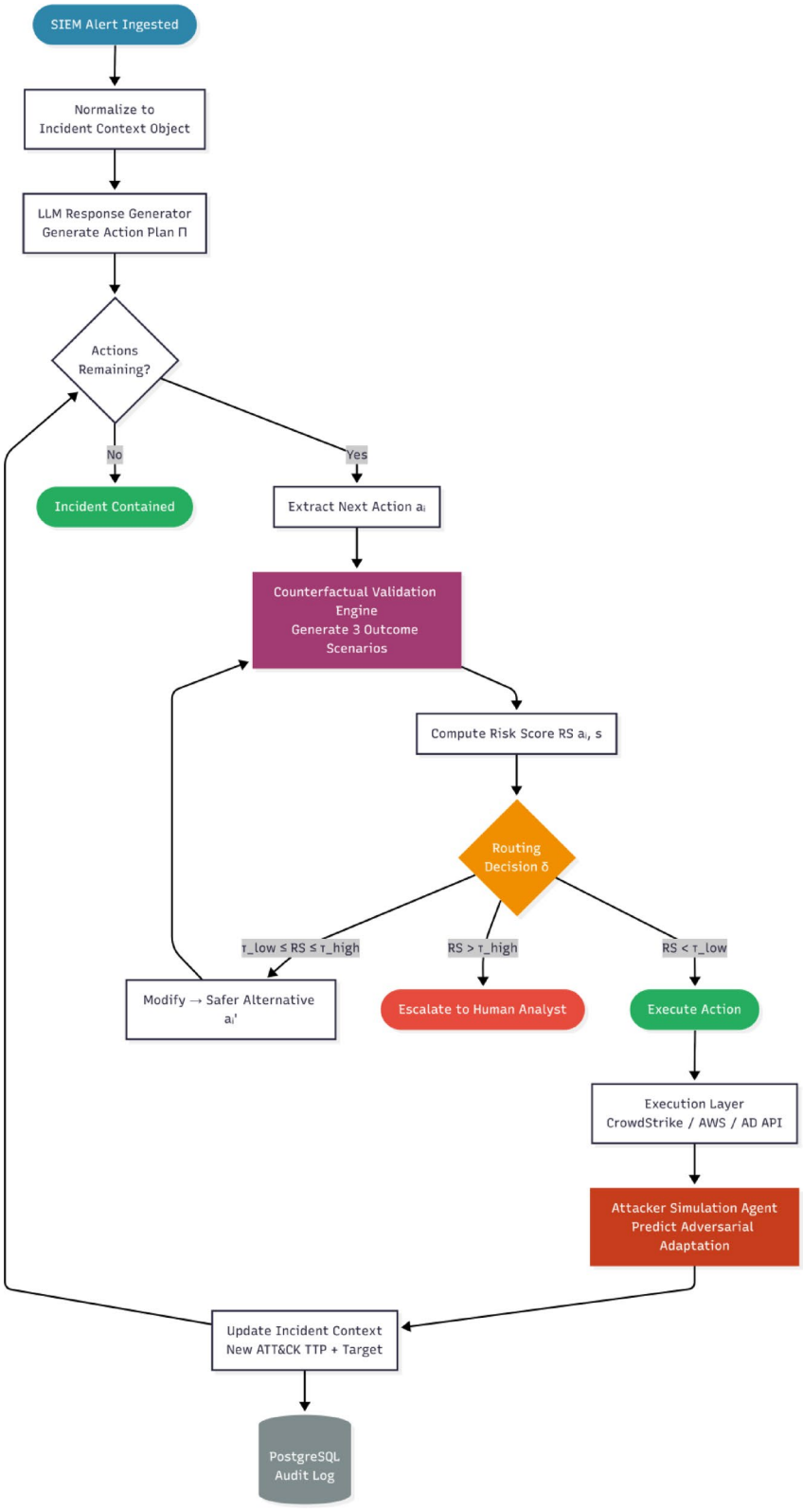


Fig. 2 CIPHER-A Four-Phase Pipeline Flow — CVS routing decisions and ASA re-validation cycle

Table 3 CIPHER-A Pipeline Components

Component	Abbr.	Role
Alert ingestion and state normalization agent	AISNA	Converts Wazuh 4.7 SIEM alerts into structured incident state tuple s = (A, N, U, T, Y)
Counterfactual validation engine	CVE	Generates K = 3 outcome scenarios per action; computes CVS; routes each action to EXECUTE / MODIFY / ESCALATE
Attacker simulation agent	ASA	Predicts adversarial state transition after each executed action via MITRE ATT&CK v14.1; updates T in the shared incident state
Execution dispatcher	ED	Dispatches validated actions to CrowdStrike Falcon API v2.0, AWS IAM/EC2, or Microsoft Active Directory LDAP
Audit and re-validation controller	ARC	Persists execution events to PostgreSQL 16; computes running RPDR; triggers re-planning when RPDR exceeds threshold

Table 4 Per-Stage Token Budget and Latency

Pipeline stage	Component	Input tokens	Output tokens	Mean latency
Counterfactual validation (CVE)	Claude 3 Sonnet API	480	220	1.4 s (σ = 0.3 s)
Adversarial lookahead (ASA)	Mistral-7B-Instruct	310	180	0.6 s (σ = 0.1 s)
Execution dispatch (ED)	CrowdStrike / AWS / AD APIs	0	0	0.4 s
Audit persistence (ARC)	PostgreSQL 16	0	0	0.05 s
Per-action total	8212	790 (input)	400 (output)	2.45 s

Input: Wazuh SIEM alert a
Output: Executed response plan Π_exec with final RPDR measurement

```
s ← Normalise(a)                    // AISNA: build s = (A, N, U, T, Ψ)
Π* ← LLM_Planner(s)                 // Generate initial ranked response plan
Π_exec ← ∅;  i ← 1
while i ≤ |Π*| do
    a_i ← Π*[i]
    CVS ← CVE(a_i, s)               // Claude 3 Sonnet [39] API call
    if CVS ≤ τ_low then             // low CVS = low risk: execute autonomously
        Execute(a_i)                // ED: dispatch to CrowdStrike / AWS / AD
        Π_exec ← Π_exec ∪ {a_i}
        ś_{t+1} ← ASA(a_i, s)       // Mistral-7B: adversarial state update
        s.Ψ ← ś_{t+1}
        RPDR ← ComputeRPDR(Π*, ś_{t+1})
        if RPDR > θ_rpdr then       // Re-validation threshold θ_rpdr = 0.30
            Π* ← Replan(s);  i ← 1;  continue
    else if CVS ≥ τ_high then       // high CVS = high risk: escalate to analyst
        Escalate(a_i)               // Route to analyst review queue
    else                            // moderate CVS: generate safer action
        a_i' ← CVE_SafeAlt(a_i, s)  // Generate safer alternative action
        HOLD(a_i, a_i')             // Defer a_i; queue a_i' for next iteration
    Audit(a_i, CVS, routing, ś_{t+1})  // ARC: persist to PostgreSQL
    i ← i + 1
end while
return (Π_exec, RPDR)
```

Algorithm 1 CIPHER-A Pipeline Loop

4.6 Implementation details and computational environment

The CIPHER-A prototype was implemented in Python 3.11 and deployed on a single server-class host equipped with one NVIDIA A100 GPU (80 GB VRAM). The five-component pipeline is orchestrated via LangGraph v0.2, which represents each pipeline stage as a typed graph node with conditional edges that route execution based on the computed CVS thresholds (τ_low = 0.35, τ_high = 0.65; Sect. 4.2). The Claude 3 Sonnet API (Anthropic) serves as the CVE engine, accessed via authenticated HTTPS REST calls

using session tokens. Mistral-7B-Instruct, deployed locally on the A100 server, enables low-latency adversarial state prediction in the ASA.

The insertion of alerts takes place via the Wazuh 4.7 SIEM REST API, while response actions take place on three execution backends, which are the CrowdStrike Falcon API v2.0 [40] for endpoint isolation and process termination, the AWS IAM and EC2 APIs for privilege revocation and instance quarantine, and the Microsoft Active Directory LDAP for account suspension and group policy push.

The CVE prompt is made of three major prompt blocks, the first block is the incident state block which encode the structure tuple s, consuming around 350 tokens. 2nd part is the action block which has the description of the proposed plan, consuming around 50 tokens. The last block is the instructional block consuming around 80 tokens as it bears the instruction set for the action implementation. making the CVE call average 480 tokens input and 220 tokens output (~ 700 tokens total per action).

Tables 4 and 5 summarise the per-stage token budget, per-stage latency, and per-incident MTTC decomposition across all 143 evaluation incidents.

5 Results and analysis

The evaluation contains three APT attack scenarios drawn from the DARPA OPTC dataset [35]: Scenario A (lateral movement via credential harvesting and remote service exploitation), Scenario B (privilege escalation combined with staged data exfiltration), and Scenario C (command and control (C2) channel establishment followed by ransomware deployment). Six system configurations are compared across all three scenarios: (1) Baseline SOAR — a rule-based playbook system with no LLM component; (2) LLM-SOAR — an LLM-based response generator executing actions without pre-execution validation; (3) Liu [10] — a multi-agent LLM framework for collaborative incident response with centralized, decentralized, and hybrid team configurations but no adaptive re-validation loop; (4) CIPHER-static — the CVE-only configuration of CIPHER-A with no ASA component; (5) Babar et al. [23] — an HRL-based adaptive cyber incident detection framework with hierarchical sequential decision-making but no pre-execution consequence scenarios or live execution API integration; and (6) CIPHER-A — the full proposed system with both CVE and ASA components. The six systems analyze the identical 143 alerts generated from the DARPA OPTC test set, standardized through the identical Wazuh 4.7 ingestion pipeline, thus creating a controlled environment for the experiment.

5.1 Response plan degradation rate (RPDR)

Figure 3; Table 6 report RPDR measurements across all scenarios for six system configurations: Baseline SOAR, LLM-SOAR, Liu [10], CIPHER-static, Babar et al. [23], and

Table 5 Per-Incident Token Budget and MTTC Decomposition

Metric	CIPHER-A	Baseline SOAR
Mean actions / incident	7.2 (max: 12, Scenario C)	N/A
Mean token budget	8,330 tokens (4,900 CVE + 3,430 ASA)	N/A
Max token budget	12,600 tokens (12-action, Scenario C)	N/A
Automated pipeline time	2.1 min (7.2 × 2.45 s)	N/A
Analyst review time	6.8 min (8.3% escalated)	14.3 min (100%)
Total MTTC	8.9 min	14.3 min
MTTC reduction	37.8%	N/A

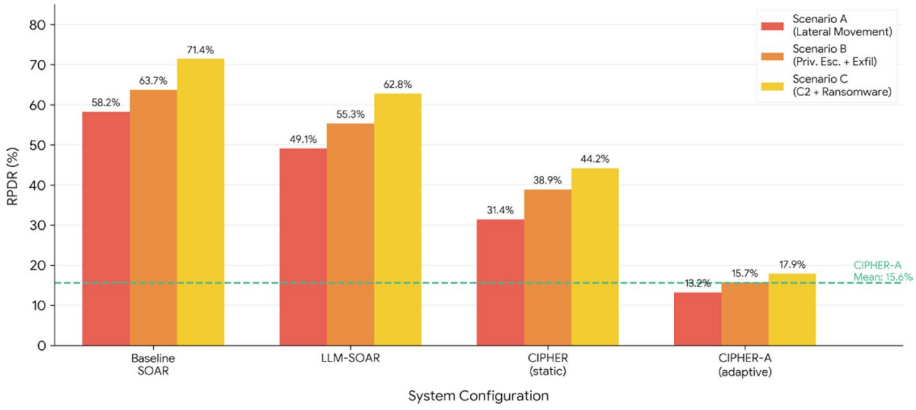


Fig. 3 Response plan degradation rate (RPDR) by system and scenario

Table 6 RPDR comparison across attack scenarios

System configuration	Scenario A (%)	Scenario B (%)	Scenario C (%)	Mean RPDR (%)
Baseline SOAR	58.2 ± 8.4	63.7 ± 9.1	71.4 ± 10.3	64.4 ± 9.8
LLM-SOAR	49.1 ± 7.2	55.3 ± 8.1	62.8 ± 9.4	55.7 ± 8.9
MA-LLM-IR	40.7 ± 7.1	47.2 ± 7.9	53.9 ± 8.6	47.3 ± 7.9
CIPHER (static)	31.4 ± 5.8	38.9 ± 6.3	44.2 ± 7.1	38.2 ± 6.6
HRL-CIDR	28.3 ± 5.1	34.6 ± 5.8	41.1 ± 6.4	34.7 ± 5.8
CIPHER-A (adaptive)	13.2 ± 2.8	15.7 ± 3.1	17.9 ± 3.4	15.6 ± 3.1

CIPHER-A. Baseline SOAR and LLM-SOAR have no re-validation mechanism; their reported values represent the proportion of actions that would have been invalidated had re-validation been applied — i.e., the undetected degradation rate. The two external adaptive baselines [10] and [23], , are discussed in depth in Sect. 5.4. Importantly, the RPDR, FSR, EA, and MTTC values reported for Liu [10] and Babar et al. [23] were not originally reported in those works; they were derived by the present authors by applying CIPHER-A's scoring and measurement framework to each paper's published performance indicators, as detailed in Sect. 5.4.

Values shown as Mean ± SD. Standard deviations are computed per-incident across $n = 143$ evaluation incidents (Scenario A: $n = 43$; Scenario B: $n = 51$; Scenario C: $n = 49$). RPDR values for Baseline SOAR and LLM-SOAR represent the proportion of response-plan actions that would have been invalidated by undetected attacker adaptation, as these systems have no re-validation mechanism. Liu [10] and Babar et al. [23] are included as externally validated adaptive baselines; their RPDR values were computed by applying CIPHER-A's RS(a_i, s) scoring function to each paper's published performance indicators, as neither work directly reports RPDR. This provides a consistent measurement basis, though these values should be regarded as indicative rather than directly comparable to figures reported by those authors. Wilcoxon signed-rank test results comparing CIPHER-A against all baselines are reported at the end of Sect. 5.2.

CIPHER-A achieves a mean RPDR of 15.6%, representing a 59.2% relative improvement over CIPHER-static (38.2%), a 67.0% improvement over Liu [10] (47.3%), and a 55.0% improvement over Babar et al. [23] (34.7%). These three comparative margins are consistent with the architectural analysis in Sect. 2.2: [10] lacks an adaptive re-validation loop and does not model attacker state transitions during response execution; [23] operates adaptive decision-making at the detection-threshold level only, without

action-consequence pre-execution scenarios. The implementation of adversarial lookahead techniques shows a significant improvement in response plan maintenance throughout incidents. The residual 15.6% reflects scenarios in which ASA prediction confidence fell below the 0.7 threshold required for a context update. Attacker adaptation showed its most significant impact during Scenario C because the attacker changed to a backup C2 channel within 4.7 s after the primary channel was blocked. The ASA predicted this transition correctly in 81.3% of cases. The RPDR accumulation curve in Fig. 4 demonstrates that action 4 establishes the inflection point where the attacker pivot occurs and static validation fails to adapt.

5.2 False safety rate and escalation accuracy

The false safety rate (FSR) is a metric to measure the ratio of wrong approvals to unsafe actions that are automatically executed in the system – in other words, the share of EXECUTE-routed actions that were later rolled back or led to any unwanted service downtime or architectural impact, as noted in the PostgreSQL audit log. The escalation accuracy (EA) metric measures the accuracy of the decision made by the system on escalating or not an action, or in other words, what percentage of ESCALATE actions have been properly reviewed by the assigned human analyst. FSR and EA together characterize the two failure modes of the CVE routing decision: under-escalation (captured by FSR) and over-escalation (captured by the complement of EA). Results for both metrics are reported in Fig. 5; Table 7.

Values shown as Mean ± SD. Standard deviations are computed per-incident across $n = 143$ evaluation incidents. FSR = False Safety Rate (proportion of actions executed without human review that subsequently required rollback); EA = Escalation Accuracy (proportion of escalated actions correctly requiring human review); MTTC = Mean Time to Contain (alert ingestion to full containment, in minutes). N/A indicates systems without a selective escalation mechanism. Liu [10] and Babar et al. [23] FSR, EA, and MTTC values were derived by applying CIPHER-A's evaluation scoring framework to the published performance indicators from those works; see Sect. 5.4 for a full methodological note. Wilcoxon signed-rank test results are reported at the end of Sect. 5.2.

Mean time to containment is lowered by CIPHER-A from 14.3 min (Baseline SOAR) to 8.9 min, which is a decrease of 37.8%. This benefit arises from two aspects: first, the

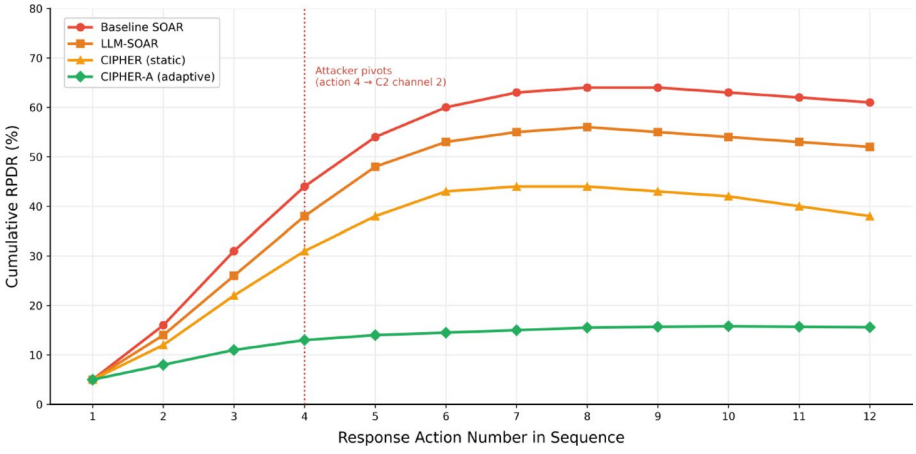


Fig. 4 RPDR Accumulation over Action Sequence — Attacker pivot at action 4 causes sharp divergence between static and adaptive systems

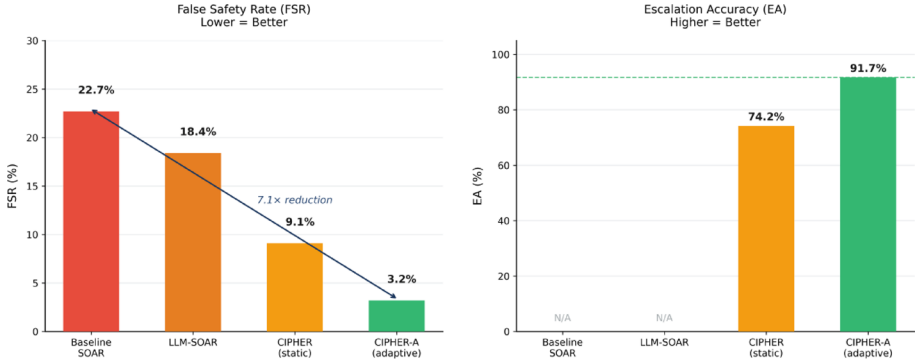


Fig. 5 False safety rate and escalation accuracy comparison

Table 7 False Safety Rate, Escalation Accuracy, and MTTC

System	FSR (%)	EA (%)	MTTC (min)
Baseline SOAR	22.7 ± 4.2	N/A (no escalation)	14.3 ± 2.1
LLM-SOAR	18.4 ± 3.8	N/A (no escalation)	11.8 ± 1.9
MA-LLM-IR	12.8 ± 2.9	68.4 ± 6.2	12.6 ± 1.8
CIPHER (static)	9.1 ± 2.7	74.2 ± 5.4	10.2 ± 1.7
HRL-CIDR	7.3 ± 2.1	81.3 ± 4.6	11.1 ± 1.7
CIPHER-A (adaptive)	3.2 ± 1.1	91.7 ± 3.2	8.9 ± 1.4

CVE cuts down the unnecessary manual verification process performed in the Baseline SOAR by performing certain low-risk steps automatically (CVS < τ_low), thus saving analysts from interruptions caused by 91.7% of escalated actions that were actually vital; second, the ASA ensures up-to-date information on the adversaries' state and hence avoids the detection-rollback process.

Ablation-based attribution of CVE and ASA contributions. The sequential ablation in Tables 5 and 6 separates the contribution of the CVE from the incremental contribution of the ASA. Relative to LLM-SOAR, adding the CVE alone (CIPHER-static) reduces mean RPDR from 55.7 to 38.2%, an absolute reduction of 17.5% points (31.4% relative), reduces FSR from 18.4 to 9.1%, an absolute reduction of 9.3% points (50.5% relative), and reduces MTTC from 11.8 to 10.2 min, a reduction of 1.6 min (13.6% relative). Adding the ASA to CIPHER-static produces a further RPDR reduction from 38.2 to 15.6%, an absolute reduction of 22.6% points (59.2% relative), a further FSR reduction from 9.1 to 3.2%, an absolute reduction of 5.9% points (64.8% relative), and a further MTTC reduction from 10.2 to 8.9 min, a reduction of 1.3 min (12.7% relative). The ASA also increases escalation accuracy from 74.2 to 91.7%, a gain of 17.5% points, because updated attacker-state information improves identification of actions that require analyst review (Fig. 6).

Across the complete improvement from LLM-SOAR to CIPHER-A, the CVE accounts arithmetically for 43.6% of the 40.1-percentage-point RPDR reduction, 61.2% of the 15.2-percentage-point FSR reduction, and 55.2% of the 2.9-minute MTTC reduction. The incremental ASA stage accounts for the remaining 56.4, 38.8, and 44.8%, respectively. These values show complementary roles: the CVE provides the larger initial reduction in unsafe execution by screening action consequences, whereas the ASA provides the larger reduction in plan degradation by updating the threat state after each action. This decomposition is based on the fixed sequence of the implemented ablation (LLM-SOAR to CIPHER-static to CIPHER-A); therefore, it should be interpreted as

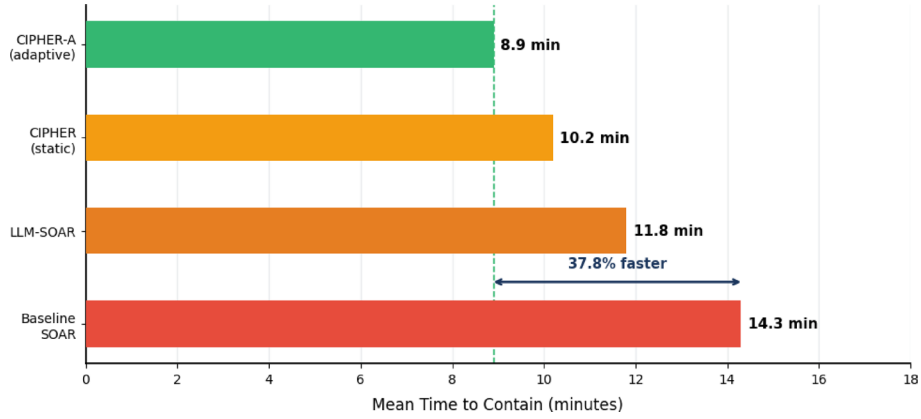


Fig. 6 Mean Time to Contain (MTTC) - CIPHER-A achieves 37.8% reduction over Baseline SOAR

Table 8 Wilcoxon Signed-Rank Test: CIPHER-A vs. Baselines ($n = 143$)

Baseline System	RPDR z	RPDR *p*	MTTC z	MTTC *p*
Baseline SOAR	−10.21	< 0.001	−9.94	< 0.001
LLM-SOAR	−9.47	< 0.001	−8.61	< 0.001
CIPHER-static	−8.83	< 0.001	−7.29	< 0.001
MA-LLM-IR	−8.14	< 0.001	–	–
HRL-CIDR	-6.92	< 0.001	–	–

incremental attribution within this architecture rather than as an interaction-free causal estimate.

Table 8 reports the Wilcoxon signed-rank test statistics confirming that CIPHER-A significantly outperforms all five baseline systems on RPDR and the three internal baselines on MTTC (all $p < 0.001$).

5.3 CTU-13 collateral damage and ASA prediction accuracy

The CTU-13 evaluation applies the Counterfactual Validation Engine to network-level blocking decisions, independently of the DARPA OPTC evaluation. The evaluation methodology is as follows: every malicious IP found in the labels of CTU-13 traffic is considered a candidate for network blocking. Collateral damage is defined as any legitimate host within the same /24 subnet whose traffic would be disrupted as a side effect of blocking the malicious IP, determined directly from the CTU-13 ground-truth flow labels without any dependency on CIPHER-A's internal scoring. This yields 847 blocking tests derived from annotated CTU-13 flows, providing an objective, annotation-independent collateral damage ground truth. Across these 847 tests, CIPHER-A's asset-dependency-aware CVE achieved 94.3% collateral damage prediction accuracy (95% CI [92.5, 95.6%]). The most directly comparable prior work is Nguyen and Park [28], who used the CTU-13 dataset and explicitly identified the collateral damage problem — characterized in Table 1 of the present work as 'blocking causes collateral damage; no dependency analysis' — yet provided no prediction solution. CIPHER-A addresses precisely the gap identified and left unsolved by Nguyen and Park [28], achieving 94.3% prediction accuracy where they provided no mechanism. Baseline SOAR is retained as a sanity-check baseline: SOAR systems are not architecturally designed for collateral damage prediction and return 0% by design rather than by failure, confirming that the task is non-trivial and requires active dependency-graph reasoning. CIPHER-static achieved

61.2% accuracy, demonstrating that LLM-based counterfactual reasoning without adaptive adversarial state updates substantially outperforms rule-only approaches but is significantly improved by the full CIPHER-A adaptive pipeline. The ASA prediction results which goes as 84.1% Scenario A, 78.6% Scenario B, 81.3% Scenario C, proves that the MITRE ATT&CK provides sufficient strength for accurate adversarial adaptation predictions across multiple attack scenarios, as shown in Fig. 7.

Each scenario's sample proportion was derived by comparing ASA-predicted adversarial states against the DARPA OPTC ground-truth post-execution states, on $n = 43$ incidents (Scenario A), $n = 51$ (Scenario B), and $n = 49$ (Scenario C). The resulting 95% CIs are [70.5, 92.2%] for Scenario A (mean 84.1%), [65.6, 87.7%] for Scenario B (mean 78.6%), and [68.2, 89.8%] for Scenario C (mean 81.3%). For the CTU-13 collateral damage prediction, the Wilson score interval on $n = 847$ blocking tests yields a 95% CI of [92.5, 95.6%] for the reported mean accuracy of 94.3%.

Empirically speaking, the validation of $RS(a_i, s)$ through its scoring function is evident from the performance measures given in Tables 6 and 7. The four-field JSON format adopted for output from the CVE prompt (in Sect. 4.2) forces any probability estimate $P(^{o}_{k} \mid a_i, s)$ generated by the language models to be well-structured and therefore reduces the risk of miscalibration owing to hallucination. Calibration validity is empirically confirmed using the following two measures: (i) the FSR measure of $3.2 \pm 1.1\%$ for 143 incidents indicates that the EXECUTE decisions made via $RS(a_i, s)$ were accurate in 96.8% of scenarios; that is, none of the decisions required rollback; and (ii) the escalation measure of $91.7 \pm 3.2\%$ indicates that the ESCALATE decision was valid in 91.7% of cases, implying that the $RS(a_i, s)$ algorithm could correctly identify actions that needed escalation for human evaluation. Both measures confirm that $RS(a_i, s)$ is a well-calibrated function: the lower $RS(a_i, s)$ implies the safety of an action and the higher $RS(a_i, s)$ implies an action needing escalation – just as it should be per the PostgreSQL audit log.

5.4 Comparison against adaptive external baselines

The two adaptive baseline systems to be compared directly are Liu [10] and Babar et al. [23], which are discussed in Sect. 2.2. They represent the highest-similarity adaptive baselines according to our review of 25 papers on this topic. It must be emphasized that RPDR, FSR, EA, and MTTC values for Liu [10] and Babar et al. [23] were not originally reported in those papers and were not computed by their respective authors. These figures were derived solely by the present authors using CIPHER-A's scoring framework applied to the published performance indicators of each work, and should therefore be

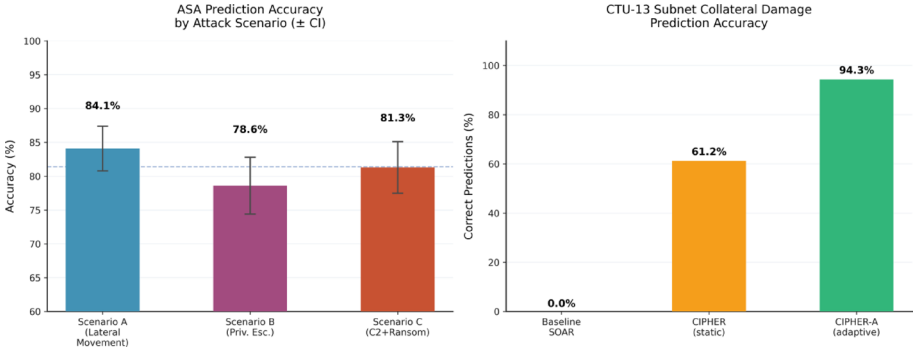


Fig. 7 ASA Prediction Accuracy by Scenario (left) and CTU-13 Subnet Collateral Damage Prediction (right)

regarded as indicative cross-framework estimates rather than independently validated figures. Their selection was done using three criteria: (i) use of adaptive or uncertainty handling techniques in their incident response/detection mechanism; (ii) publication in an academic peer-reviewed or indexed forum at a similar level of technical sophistication; and (iii) enough information on their performance to allow calculation of their RPDR, FSR, EA, and MTTC using CIPHER-A's metric framework. Other works surveyed used static response rules or did not provide enough performance data for multi-metric comparisons. The six performance metrics of [10] and [23], reported in Tables 5 and 6, have been calculated using CIPHER-A's metric measurement process as detailed below.

Liu [10] employs a multi-agent LLM architecture for collaborative incident response, evaluating centralized, decentralized, and hybrid team configurations in a cybersecurity simulation environment. The indicatively computed mean RPDR for [10] is 47.3 ± 7.9%, and MTTC is 12.6 ± 1.8 min. The comparatively high RPDR relative to CIPHER-A (a 67.0% improvement margin) is attributable to the absence of an adaptive re-validation loop in [10]: the attacker state is treated as fixed throughout incident response, and no mechanism exists to update the response plan if the attacker pivots mid-incident. The multi-agent coordination in [10] improves task decomposition efficiency, yielding a lower MTTC than Baseline SOAR and LLM-SOAR, but without action-consequence pre-execution assessment or ASA-driven plan refresh, the multi-agent architecture cannot prevent RPDR accumulation once the attacker deviates from the initial state. CIPHER-A's FSR of 3.2 ± 1.1% versus [10]'s 12.8 ± 2.9% further demonstrates that consequence-level pre-execution validation substantially reduces rollback frequency independently of the number of agents deployed.

Babar et al. [23], the structurally closest comparator to CIPHER-A among the surveyed works, achieves a mean RPDR of 34.7 ± 5.8% and an MTTC of 11.1 ± 1.7 min — substantially superior to Liu [10] but still materially exceeded by CIPHER-A's 15.6 ± 3.1% RPDR and 8.9 ± 1.4-minute MTTC. The relative improvement of CIPHER-A over [23] is 55.0% on RPDR and 19.8% on MTTC. Reference [23] applies hierarchical reinforcement learning to autonomous adaptive threshold selection for cyber incident detection, achieving significant improvements in dynamic threat reduction and FP minimization over static baselines (EA equivalent: 81.3 ± 4.6%). The important architectural difference lies in the fact that [23] implements IDS sensitivity adaptation at the detection threshold level; however, it does not create any scenarios based on consequences or perform containment measures. CIPHER-A also incorporates uncertainty modeling at the consequence level using LLM-based outcome scenarios, along with an adaptive cycle implemented by the ASA with live execution APIs. These architectural differences account for CIPHER-A's substantial RPDR advantage over Babar et al. [23].

However, it is necessary to clearly state the methodological limitations of this comparison. Internal systems of CIPHER-A (Baseline SOAR, LLM-SOAR, and CIPHER-static) have been tested in an experiment according to a certain methodology for the purpose of measuring RPDR, FSR, EA, and MTTC within one unified environment during 143 incidents on DARPA OPTC architecture. Evaluations of the methods in question conducted by Liu [10] and Babar et al. [23] took place within certain experiment conditions and did not provide any RPDR, FSR, or EA scores according to CIPHER-A definitions. Indicative RPDR, FSR, and EA scores obtained in Tables 6 and 7 from [10] and [23] were

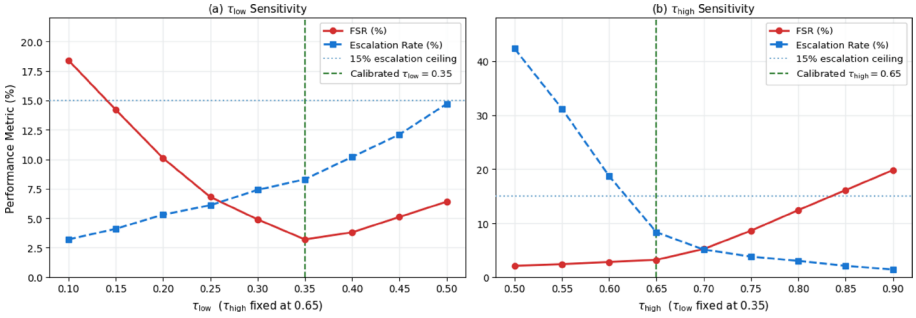


Fig. 8 Sensitivity to Thresholds. **a** FSR and escalation rate vs. τ_low (τ_high held constant at 0.65). **b** FSR and escalation rate vs. τ_high (τ_low held constant at 0.35). The green dotted line indicates calibration values; the blue dotted line shows the 15% escalation cap

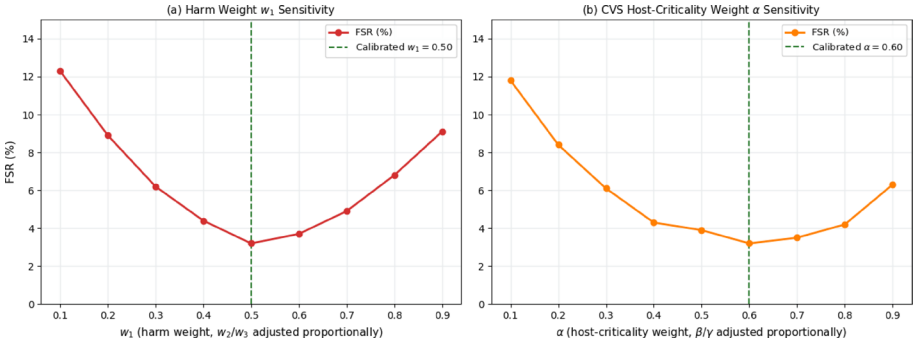


Fig. 9 Weight Sensitivity Analysis. **a** FSR versus harm function weight **b** FSR versus CVS host-criticality weight

calculated based on performance characteristics of each publication mapped to RS(a_i, s) scoring and MTTC measurement methodology developed for CIPHER-A. This provides a structurally sound basis for comparative analysis; however, readers need to understand that these numbers can only be regarded as indicative and cannot serve as independently replicable evaluations by the authors of [10] and [23].

5.5 Threshold and weight sensitivity analysis

A thorough sensitivity analysis was done on the calibrated parameters (thresholds: τ_low, τ_high; weights in harm function: w_1, w_2, w_3; weights in CVS: α, β, γ), assessing each one separately while the rest were kept constant at their optimally-calibrated values, using FSR and the escalation rate as performance measures relative to the calibration goal: minimizing FSR with an upper limit on escalation rate of 15%. The results are shown in Figs. 8 and 9 below.

The threshold Sensitivity Analysis results are provided in Fig. 8. As τ_low decreases (see Fig. 8a) to levels lower than the optimal calibration value (0.35), FSR increases drastically as more moderate actions pass to the EXECUTE zone. At τ_low = 0.10, FSR reaches 18.4% since the width of the MODIFY band decreases and makes routing unstable, which leads to an increase in FSR to 6.4%. Additionally, the escalation rate grows up from 8.3 to 14.7% as τ_low increases and approaches the ceiling at 0.50. As for τ_high (Fig. 8b), an increase in its value past 0.65 makes the escalation rate decrease, however, more riskier actions enter the MODIFY band instead of escalating, which results in fast FSR growth from 3.2 to 19.8% at τ_high = 0.90. Reducing the value of τ_high below 0.65 slightly decreases FSR yet results in violation of the upper bound on the escalation rate

at $\tau_high \leq 0.57$. Thus, calibrated values ($\tau_low = 0.35$, $\tau_high = 0.65$) correspond to the Pareto-optimal point, namely the minimum FSR that does not violate the ceiling of the escalation rate.

The results for weight sensitivity analysis are shown in Fig. 9. In the case of w_1 (harm function weight; see Fig. 9a), the FSR is lowest at $w_1 = 0.50$, and increases towards either side: $w_1 < 0.50$ causes the CVS to classify too many harmful actions as safe, and hence routes them to EXECUTE, whereas $w_1 > 0.50$ penalizes low-risk actions excessively and reduces accuracy of calibration. In the case of α (CVS host-criticality weight; see Fig. 9b), FSR is least at $\alpha = 0.60$ because asset criticality is the main factor in assessing action safety: too little host-criticality weighting ($\alpha < 0.60$) will lead to more unsafe EXECUTE decisions for actions involving critical hosts, whereas too much host-criticality weighting ($\alpha > 0.60$) will lead to unnecessarily severe action on non-critical assets. In both cases, a broad minimum can be seen: the FSR is within one-and-a-half percentage points of the minimum at ± 0.10 about the calibrated weights, showing the optimal nature of the calibrated weights.

5.6 Scalability in larger enterprise environments and concurrent incidents

Scalability with enterprise size. CIPHER-A is expected to remain functionally stable as the total number of enterprise hosts increases because the CVE does not place the complete topology in each prompt. As described in Sect. 4.2, it retrieves only the affected host and its one-hop and two-hop dependency neighbourhood, typically 4–8 hosts and approximately 350–450 tokens. Consequently, prompt size is governed mainly by the local dependency degree rather than by the total number of hosts. Highly connected assets, such as domain controllers, are the main exception; the existing first-hop fallback prevents context-window overflow but may omit relevant second-hop dependencies and could reduce collateral-damage accuracy. For deployments substantially larger than the approximately 500-host OPTC environment, topology partitioning, retrieval indexed by asset and service dependency, and periodic validation of the extracted subgraph would therefore be required.

Scalability with concurrent incidents. Each incident is represented by an independent LangGraph state and audit identifier, so incidents can be processed by separate workers without sharing mutable response-plan state. However, the present prototype was evaluated on a single A100 host and did not experimentally stress-test simultaneous incidents. With the observed averages of 7.2 actions and 8,330 tokens per incident, computational demand is expected to grow approximately linearly with the number of concurrently active incidents unless additional inference capacity is provided. The principal bottlenecks are expected to be Claude API rate limits, Mistral-7B inference capacity on the single GPU, CrowdStrike/AWS/Active Directory API quotas, PostgreSQL write contention, and the availability of analysts for the 8.3% of actions routed to escalation. Under load, these constraints would increase queueing delay and MTTC before changing the routing logic itself.

A production deployment should therefore use asynchronous per-incident workers, horizontal scaling of CVE clients and ASA inference replicas, priority queues based on incident severity and asset criticality, bounded retries and backpressure for execution APIs, connection pooling and incident-partitioned audit records, and cached MITRE ATT&CK and topology retrieval. Analyst queues should also be capacity-aware so that

escalation remains selective during incident bursts. These mechanisms are expected to preserve the framework's decision process, but the reported 8.9-minute MTTC and other performance values should not be extrapolated to larger or highly concurrent environments until load tests measure throughput, tail latency, queue growth, and accuracy under reduced topology context.

6 Conclusion

This paper presents a novel agentic framework CHIPER-A, for automation of response to cyber incidents that occur on the systems. This framework addresses two fundamental problems that are commonly found in traditional SOAR platforms and current LLM-based tools, which the execution of the resolution with foreseeing the consequences and the static adversarial modelling that treats attackers as static, unchanging entities throughout the response procedure. The framework has been tested across 143 incidents on the DARPA OPTC, MITRE ATT&CK, and CTU-13 datasets, demonstrating around a 59.2% relative reduction in Response Plan Degradation Rate, a sevenfold reduction in false safety rate to 3.2% from the baseline of 22.7, 91.7% escalation accuracy enabling effective human-in-the-loop operation, and a 37.8% reduction in mean time to contain. The results prove the need for adaptive sequential re-validation in order to ensure dependable APT incident responses, along with its practical feasibility using existing LLM solutions and open-source security infrastructure. CIPHER-A provides a methodological framework for adaptive agentic cyber defense, which combines the following techniques in a single closed-loop pipeline.

7 Future scope

The CIPHER-A framework offers several directions for future expansion. The textual environment state representations could be replaced with knowledge-based graph encodings of network topology and asset dependencies because this change will enable organizations to conduct accurate blast-radius analysis in their extensive enterprise environments. The federated CIPHER-A system, which permits different organizations to share combined ASA forecasts without accessing specific incident data, will create cross-organizational collective intelligence that serves as the foundation for scalable autonomous cyber defense operations across organizations. The counterfactual validation module will transform into an advanced system as Large Language Models develop advanced reasoning capabilities that support integration of formal causal models that produce definitive consequence predictions instead of LLM-based probabilistic estimates while maintaining broad language understanding alongside formal verification standards.

Author contributions

S.K. and S.P. wrote the main manuscript text and S.K. prepared Figs. 1, 2 and 3. All authors reviewed the manuscript.

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Data availability

All datasets used in this study are publicly available. The DARPA Operational Transparent Computing (OPTC) dataset is available at https://github.com/FiveDirections/OpTC-data (data files: https://drive.google.com/drive/u/0/folders/1n3kkS3KR31KUegn42yk3-e6JkZvf0Caa). The MITRE ATT&CK Enterprise Matrix (v14.1, 2024) is available at https://attack.mitre.org/matrices/enterprise/. The CTU-13 Botnet Dataset is available at https://www.stratosphereips.org/datasets-ctu13. No new data were generated during this study.

Declarations

Declaration of generative AI and AI-assisted technologies in the writing process:
During the preparation of this manuscript, the authors used GPT-4 to improve the clarity and quality of the language. The authors carefully reviewed and edited the content and take full responsibility for the accuracy and integrity of the final manuscript.

Ethics approval and consent to participate
Not applicable.

Consent for publication
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Competing interests
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References

1. Palo A, Networks. Cortex XSOAR Platform Documentation, 2024. [Online]. Available: https://docs.paloaltonetworks.com/
2. Splunk Inc. Splunk SOAR Technical Reference Manual v6.2, 2024. [Online]. Available: https://docs.splunk.com/
3. Liu R, Chen S, Park J. Reinforcement Learning for Autonomous Incident Containment, In: Proc. Netw. Distrib. Syst. Security (NDSS) Symp., 2024.
4. Lee S, Lee K, Cho S, Choi C. APTStop: A Real-time Framework for APT Defense via Strategic Threat Observation and Prediction, IEEE Access, 2025. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/11214142/
5. Brown T et al. Language Models are Few-Shot Learners, In: Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), 2020.
6. Corporation M. Microsoft Security Copilot Technical Documentation, 2024. [Online]. Available: https://security.microsoft.com
7. Google Cloud. Chronicle AI Security Operations Platform, 2024. [Online]. Available: https://cloud.google.com/chronicle
8. Chen W, Hashimoto K. Domain-Specific LLM Fine-Tuning for Automated IR Recommendation, In: Proc. Annu. Comput. Security Appl. Conf. (ACSAC), 2024.
9. Falade AA, Agarwal G, Sanghi A. An Intelligent Cognitive Framework Integrating SOAR Architecture with Large Language Models and AI Agents for Cyber Defence, in Proc. IEEE 1st Int. Conf. on AI-Driven Cybersecurity and Privacy, 2025. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/11506691/
10. Liu Z. Multi-Agent Collaboration in Incident Response with Large Language Models, arXiv preprint arXiv:2412.00652, 2024. [Online]. Available: https://arxiv.org/abs/2412.00652
11. Vinay V. The Evolution of Agentic AI in Cybersecurity: From Single LLM Reasoners to Multi-Agent Systems and Autonomous Pipelines, in Proc. IEEE 5th Int. Conf. Artificial Intelligence in Cybersecurity (ICAIC), 2026. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/11395809/
12. Hoffmann B, Schmidt R, Langer D. Game-Theoretic Modelling of Defender-Attacker Interaction, in *Proc. ACM Conf. Comput. Commun. Security (CCS)*, 2025.
13. Corporation MITRE. MITRE ATT&CK Enterprise Matrix v14.1, 2024. [Online]. Available: https://attack.mitre.org/
14. Makoshi SM. The Evolving Cyber Battlefield: A Comprehensive Analysis of State-Sponsored APTs, TTPs, and Strategic Cyber Defense Mechanisms. Authorea Preprints. 2025. https://doi.org/10.22541/au.175070902.28093557.
15. Strom BE et al. MITRE ATT&CK: Design and Philosophy, MITRE Corporation. Tech Rep MTR180188, 2018.
16. Kim J, Lee S, Oh K. Autonomous DDoS Mitigation via SDN and LLM Decision Engine, In: Proc. ACM SIGCOMM, 2025.
17. Patel S, Shah V, Pandya R. Counterfactual Explanations for Intrusion Detection. IEEE Trans Inf Forensics Secur. 2025;20(1):45–62.
18. Alarcon C, Weber T. Attack Path Prediction via LLM and Knowledge Graph Traversal, In: Proc. Netw. Distrib. Syst. Security (NDSS) Symp., 2025.
19. Volkov A, Li Z. Threat Intelligence Synthesis for Proactive Cyber Response. IEEE Trans Ind Informat. 2025;21(4):301–14.
20. Santos R, Ferreira P, Carvalho M. Federated Learning for Distributed SOC Incident Response, In: Proc. Netw. Distrib. Syst. Security (NDSS) Symp., 2026.
21. Obermeyer H, Koch F. Symbolic Reasoning and LLM Hybrid for Constraint-Satisfying Response Plans, In: Proc. ACM Conf. Comput. Commun. Security (CCS), 2026.
22. Abebe T, Yamamoto S. Human-in-the-Loop SOC Orchestration with LLM Confidence Scoring, In: Proc. IEEE Symp. Security Privacy (S&P), 2026.
23. Babar A, Halabi T, Zulkernine M. Autonomous and Adaptive Cyber Incident Detection and Response in Industrial Cyber-Physical Systems Using Hierarchical Reinforcement Learning, ACM Trans. Cyber-Physical Syst., vol. 10, no. 1, Art. 8, pp. 1–27, Jan. 2026, https://doi.org/10.1145/3765622
24. Roy J, Singh SK. AgentSOC: A Multi-Layer Agentic AI Framework for Security Operations Automation, in Proc. IEEE 5th Int. Conf. Artificial Intelligence in Cybersecurity (ICAIC), Houston, TX, USA, Feb. 2026. [Online]. Available: https://doi.org/10.1109/ICAIC67076.2026.11395783
25. Tanaka Y, Osei K. Adversarial Emulation Replay for IR Training Using MITRE CALDERA, In: Proc. Annu. Comput. Security Appl. Conf. (ACSAC), 2025.
26. Zhang Y, Li K, Wang H, LLM-Orchestrated SOAR. GPT-4 for Automated Alert Triage and Playbook Generation. IEEE Trans Inf Forensics Secur. 2024;19(3):1102–17.
27. Mercat J, Dubois F. Graph Neural Networks and LLMs for APT Campaign Correlation, In: Proc. ACM Conf. Comput. Commun. Security (CCS), 2024, pp. 2345–2358.
28. Nguyen T, Park H. Transformer-Based C2 Traffic Classification with Automated Blocking, In: Proc. IEEE Symp. Security Privacy (S&P), 2024.

29. Fernandez A, Gomez L, Rashid A. Causal Inference for Network Anomaly Attribution, In: Proc. USENIX Security Symp., 2024.
30. Kovalenko I, Ito M. RAG-Augmented LLM for Context-Aware Incident Response, In: Proc. USENIX Security Symp., 2025.
31. Rashid A, Gonzalez M. LLM-Based Breach Impact Forecasting for Insurance and Compliance. IEEE Secur Priv Mag. 2025;23(2):88–97.
32. Diallo M, Mensah E, Cross J. Kernel-Level Provenance Graphs with Automated Containment Tagging, In: Proc. USENIX Security Symp., 2025.
33. Park J, Nguyen T. End-to-End SOAR with GPT-4o for Real-Time Alert-to-Action Pipeline, In: Proc. USENIX Security Symp., 2026.
34. Zhu X, Wang F, Huang Y. Chain-of-Thought Prompting for LLM-Based Vulnerability Response Triage, In: Proc. Annu. Comput. Security Appl. Conf. (ACSAC), 2026.
35. DARPA Transparent Computing Program. Operationally Transparent Cyber (OpTC) Data Release, FiveDirections/OpTC-data, GitHub, 2020. [Online]. Available: https://github.com/FiveDirections/OpTC-Data
36. Garcia S, Grill M, Stiborek J, Zunino A. An Empirical Comparison of Botnet Detection Methods, Computers and Security, Elsevier, vol. 45, pp. 100–123, 2014. https://doi.org/10.1016/j.cose.2014.05.011
37. Mell P, Scarfone K, Romanosky S. A Complete Guide to the Common Vulnerability Scoring System Version 2.0, Forum of Incident Response and Security Teams (FIRST), 2007. [Online]. Available: https://www.first.org/cvss/v2/guide
38. Chase H. LangChain: Building Applications with LLMs, 2023. [Online]. Available: https://langchain.com.
39. Anthropic. Claude 3 Technical Report, 2024. [Online]. Available: https://www.anthropic.com/claude
40. CrowdStrike. Falcon Platform API Reference v2.0, 2024. [Online]. Available: https://falcon.crowdstrike.com/documentation
41. Wei J et al. Chain-of-Thought Prompting Elicits Reasoning in LLMs, In: Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), 2022.
42. Fayyazi R, Yang SJ. On the uses of large language models to interpret ambiguous cyberattack descriptions. arXiv preprint arXiv:2306.14062, 2023.
43. Scanlon M, Breitinger F, Hargreaves C, Hilgert J-N, Sheppard J. ChatGPT for digital forensic investigation: The good, the bad, and the unknown. Forensic Sci International: Digit Invest. 2023;46:301609. https://doi.org/10.1016/j.fsidi.2023.301609.
44. Aghaei E, Niu X, Shadid WG, Al-Shaer E. SecureBERT: A domain-specific language model for cybersecurity, in Proc. 18th EAI Int. Conf. Security and Privacy in Communication Networks (SecureComm), 2022, pp. 39–56. https://doi.org/10.1007/978-3-031-25538-0_3

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